

Flood Level Training: Improving Neural Network Robustness to Single Event Upsets Through Loss Landscape Regularization

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Abstract

Neural networks in radiation environments (space, nuclear, accelerators) are vulnerable to Single Event Upsets (SEUs): transient bit flips in parameters. While hardware protections exist, they incur significant overhead. We investigate whether *flood level training*—a regularization technique preventing zero training loss—can improve inherent model robustness. In this **proof-of-concept study on small MLPs and synthetic datasets**, we explore whether flooding encourages convergence to flatter, more robust loss minima. Across 36 configurations, flood training reduces SEU vulnerability by up to 10.0% on average at $b = 0.15$ ($20.0\times$ ROI), with individual configurations showing up to $\sim 49\%$ improvement. However, the effect is dataset-dependent: flooding is only active when the flood level exceeds the model’s natural training loss. Dropout alone provides a 15.1% robustness improvement. Per-bit analysis reveals the exponent MSB (bit 1) dominates all vulnerability. While establishing feasibility, generalizability to large-scale models requires further validation.

1 Introduction

1.1 Motivation: Hardware Faults in Harsh Environments

Neural networks deployed in space missions, nuclear facilities, and particle accelerators face **Single Event Upsets (SEUs)**—transient bit flips caused by ionizing radiation. A single parameter flip can cause catastrophic failure. Traditional hardware mitigations (ECC, TMR) incur 30–300% overhead. We investigate a complementary approach: *training methodology* for inherent robustness.

1.2 Flood Level Training

Flood level training [1] prevents overfitting by maintaining a minimum loss threshold b : $\mathcal{L}_{\text{flood}}(\theta) = |\mathcal{L}(\theta) - b| + b$. **Hypothesis:** By preventing zero loss, flooding encourages flatter minima, potentially improving robustness to parameter perturbations like bit flips.

1.3 Contributions

Primary Question: Does flood level training improve SEU robustness?

Contributions:

1. First **proof-of-concept study** of flood training for SEU robustness (small MLPs, synthetic data).

2. Evidence of up to 10.0% average vulnerability reduction at $b = 0.15$ (0.50% accuracy cost, $20.0\times$ ROI), with dataset-dependent effects up to $\sim 49\%$.
3. Discovery that flooding effectiveness requires calibration: flood level must exceed natural training loss.
4. Per-bit-position analysis revealing exponent MSB (bit 1) dominates all vulnerability.
5. Public release of data and code.

Scope: This study establishes feasibility. Results on simplified benchmarks may not directly generalize to large-scale models or complex tasks (see Section 5.3).

2 Related Work

SEU Robustness: Dennis & Pope [2] established the SEU injection framework used here, showing architectural choices impact fault tolerance. We extend this to *training methodology*.

Flood Level Training: Ishida et al. [1] introduced flood training to improve generalization. It complements dropout and weight decay.

Loss Landscape: Flat minima generalize better and are less sensitive to perturbations [3, 4]. We hypothesize flooding encourages such minima, improving SEU robustness.

3 Methodology

Design: We compare standard vs. flood training across 36 configurations: 3 synthetic datasets (moons, circles, blobs; 2000 samples), 6 flood levels ($b \in [0.0, 0.05, 0.10, 0.15, 0.20, 0.30]$), and 2 dropout settings (0.0, 0.2).

Model: 3-layer MLP ($2 \rightarrow 64 \rightarrow 32 \rightarrow 1$, ReLU, 2,305 params). Trained with Adam (lr=0.01) for 100 epochs using binary cross-entropy (wrapped with flooding).

SEU Injection: Following [2], we simulate single-bit flips in float32 parameters at 5 representative bit positions: bit 0 (sign), bit 1 (exponent MSB), bit 8 (exponent LSB), bit 9 (mantissa MSB), and bit 31 (mantissa LSB). We test 15% of parameters (~ 345 injections/bit position) and measure mean accuracy drop.

4 Results

4.1 Robustness and Cost-Benefit Analysis

Flooding can reduce SEU vulnerability, but the effect is dataset-dependent and non-monotonic (Figure 1). Table 1 summarizes cross-dataset averages. The optimal cross-dataset configuration ($b = 0.15$) yields a 10.0% robustness gain for 0.50% accuracy cost, achieving a $20.0\times$ ROI (Figure 2). At $b = 0.10$, ROI is $43.0\times$ but the robustness gain is smaller (3.6%). Importantly, blobs (where flooding is strongly active) drives most of the average improvement, while circles (where training loss ~ 0.43 exceeds all flood levels) shows no benefit.

4.2 Training Dynamics

Figures 3, 4, and 5 confirm that flooding actively constrains training—but only when the flood level exceeds natural training loss. For blobs (loss ≈ 0), all flood levels are active and final losses match

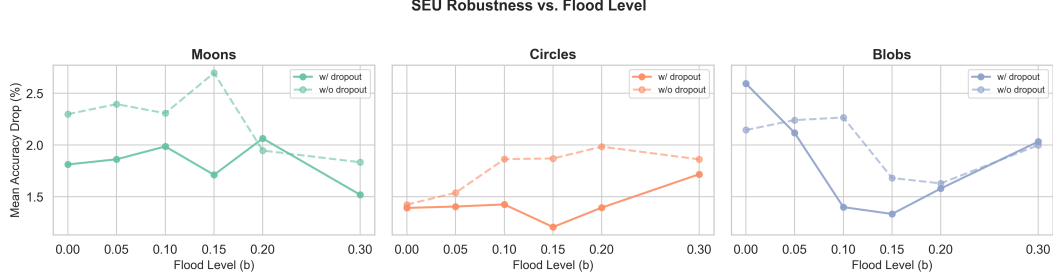


Figure 1: Mean accuracy drop under SEU injection vs. flood level. Flooding improves robustness for datasets where it is active (blobs, partially moons), but not for circles where training loss exceeds all flood levels.

Table 1: Cross-dataset average results. Best balance at $b = 0.15$; highest ROI at $b = 0.10$.

Flood Level	Baseline Acc	Acc Drop	Rel. Improvement	ROI
0.00 (std)	90.46%	1.94%	0% (baseline)	—
0.05	89.67%	1.93%	0.9%	$1.2\times$
0.10	90.38%	1.87%	3.6%	$43.0\times$
0.15	89.96%	1.75%	10.0%	20.0 $\times$
0.20	90.58%	1.77%	9.2%	N/A*
0.30	89.42%	1.83%	6.0%	$5.8\times$

* $b = 0.20$ shows negative accuracy cost (random variation), making ROI undefined.

targets. For circles (loss ≈ 0.43), no flood level is active, explaining the absence of benefit. This is a critical prerequisite for flood training effectiveness.

4.3 Consistency

The robustness improvement varies by dataset and dropout configuration (Figure 6). Blobs shows the strongest benefit (up to $\sim 49\%$ with dropout at $b = 0.15$), moons shows modest improvement, and circles shows no consistent benefit. Dropout alone provides a 15.1% robustness improvement (mean drop: 2.00% $\rightarrow$ 1.70%) with negligible accuracy cost (0.10%), making it a strong independent technique. Per-bit analysis reveals bit 1 (exponent MSB) dominates all vulnerability, with other tested bits causing near-zero accuracy drops.

5 Discussion

5.1 Mechanism: Loss Landscape Regularization

We hypothesize flooding encourages flatter loss minima. Mathematically, for parameter θ and perturbation δ , expected accuracy drop is $\approx \sum_i p(i) |\nabla_{\theta_i} \mathcal{L}| |\delta_i|$. Second-order analysis $\mathcal{L}(\theta + \delta) \approx \mathcal{L}(\theta) + \delta^T \nabla \mathcal{L} + \frac{1}{2} \delta^T H \delta$ suggests flatter minima (lower Hessian eigenvalues) reduce sensitivity [3]. Our results partially support this: for datasets where flooding is active (blobs, partially moons), robustness improves. For circles, where flooding is never active (training loss $\gg$ flood level), no

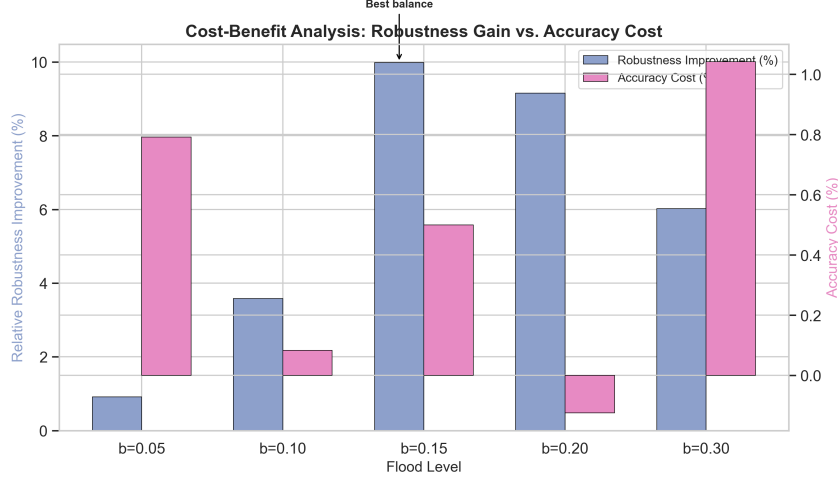


Figure 2: Cost-benefit analysis. $b = 0.15$ provides best balance; $b = 0.10$ highest ROI.

benefit is observed. This confirms that the mechanism depends on flooding actually constraining the optimization, not merely being present in the loss function.

5.2 Practical Implications

Zero Inference Overhead: Unlike ECC/TMR, flooding has no deployment cost. **Simplicity:** Easy implementation (see below) wrapping any loss function. **Compatibility:** Works with standard architectures and regularization (dropout). **Cost-Effective:** Ideal for resource-constrained environments (space, edge). **Prerequisite:** Flood level must exceed natural training loss to be active. **Dropout:** Provides 15.1% independent robustness improvement; always recommended.

```
class FloodingLoss(nn.Module):
    def __init__(self, base_loss, flood_level=0.10): ...
    def forward(self, preds, targets):
        return torch.abs(self.base_loss(preds, targets) - self.flood_level)
            + self.flood_level
```

5.3 Limitations and Threats to Validity

Scale & Generalizability: Results on small MLPs and synthetic data may not transfer to large models (ResNets, Transformers) or complex tasks (ImageNet). Different architectures (CNNs, attention) may behave differently. **Threat Model:** Single-bit flips simplify real radiation effects (multi-bit, permanent faults, latch-ups). **Theory:** Loss curvature (Hessian) was not directly measured.

6 Conclusion

This proof-of-concept study demonstrates that flood level training can improve SEU robustness on simplified benchmarks, but the effect is dataset-dependent and non-monotonic. Optimal $b = 0.15$ yields 10.0% average improvement ($20.0\times$ ROI), with individual configurations reaching $\sim 49\%$. Critically, flooding is only effective when the flood level exceeds the model’s natural training loss.

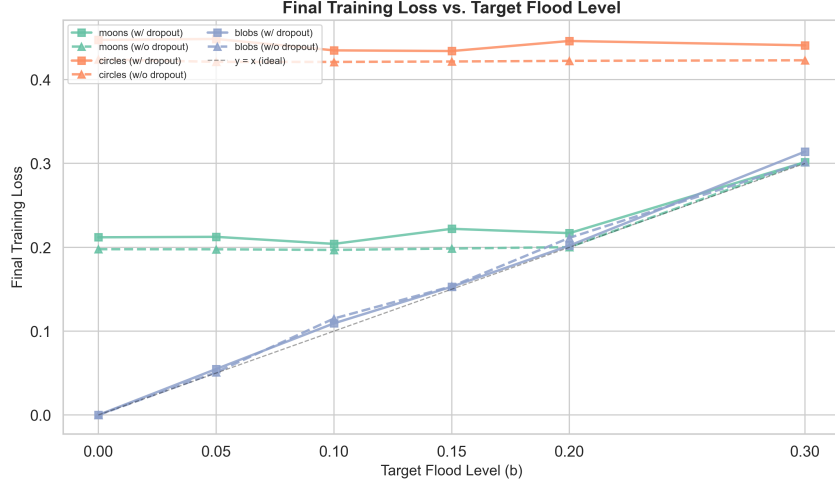


Figure 3: Final training loss vs. target flood level.

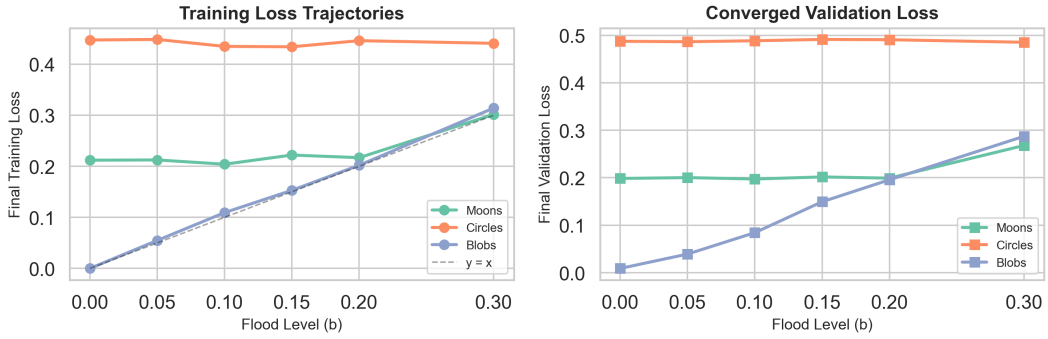


Figure 4: (Left) Training loss trajectories. (Right) Final converged loss vs. flood level.

Dropout alone provides 15.1% robustness improvement and should always be used. Per-bit analysis reveals the exponent MSB (bit 1) dominates all vulnerability, suggesting targeted hardware protection. While promising as a zero-overhead, training-time intervention, critical validation is needed on large-scale models and real hardware.

6.1 Future Work

1. **Scale-up:** Validate on CNNs (CIFAR/ImageNet) and Transformers.
2. **Architecture:** Test convolutional, attention, and normalization layers.
3. **Theory & Hardware:** Measure Hessian spectra; validate with FPGA/beam testing.
4. **Threats:** Extend to multi-bit and permanent faults.

Comprehensive Training Dynamics Analysis

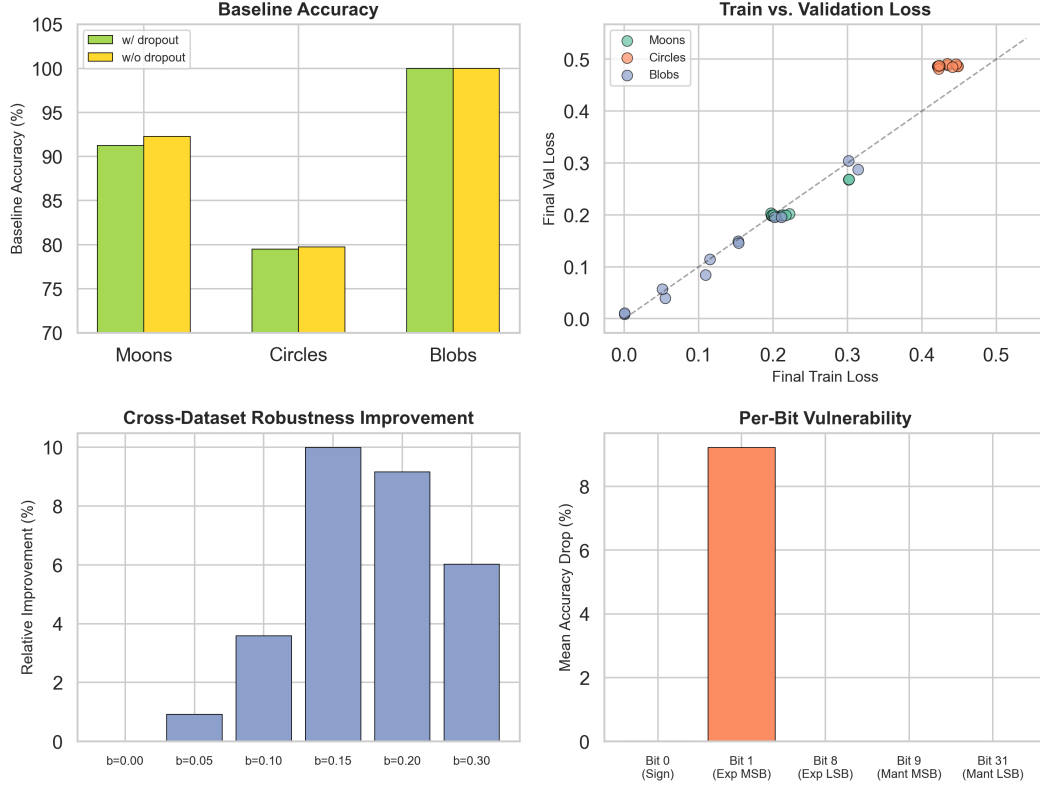


Figure 5: Comprehensive training dynamics analysis: validation accuracy, loss comparison, gradient norms, and robustness summary.

6.2 Data Availability

Code and data are available at: https://github.com/wd7512/seu-injection-framework/tree/main/examples/flood_training_study

Acknowledgments

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References

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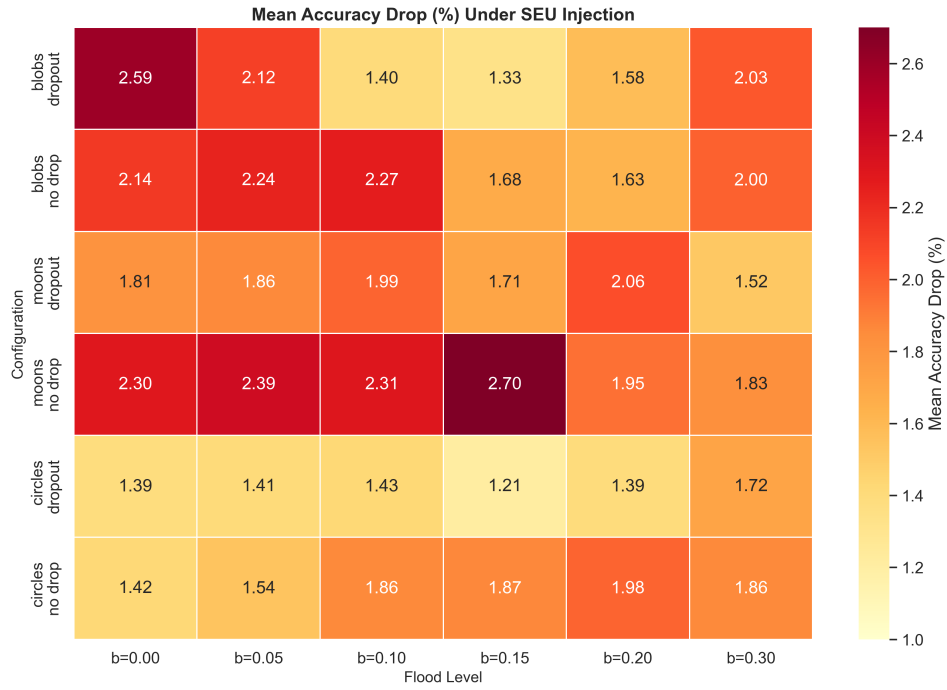


Figure 6: Heatmap of mean accuracy drop (%) across all configurations. Blobs shows strongest flooding benefit; circles shows no consistent improvement.

- [3] Sepp Hochreiter and Jürgen Schmidhuber. Flat minima. *Neural Computation*, 9(1):1–42, 1997.
- [4] Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations (ICLR)*, 2017.